Progress report

Project: Auditable AI Agents
As of: 2026-10-30
Repository: `/Users/mani-2209/AuditableAgentsProject` (local git, no remote)

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1. Work log

Step	What was done	Commit
1	Project chosen after comparing three ideas (audit trail, registry and permissions, task escrow); scaffold, proposal, locked decisions, backlog	Initial scaffold
2	Hash chain, Merkle tree with proofs, canonical JSON; first 18 tests	same
3	AuditAnchor.sol contract; Hardhat set up; Python-generated vectors verified on-chain	Add Hardhat setup...
4	Demo agent with 3 tools; audit logger; batch planning; verifier; web3 client; end-to-end demo with tamper scenarios	Add demo agent, audit logger...
5	Claude client (tested with a fake SDK, not run live: no API key in the organisation setup)	Add Claude LLM client...
6	Local LLM client for Ollama; Ollama installed, qwen3:8b downloaded (5.2 GB); real agent run logged, anchored and verified	Add local LLM client...
7	Complete test suite; log-verification script; documentation set and PDFs	current

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2. Current numbers

Item	Value
Python tests	79 passing
Contract tests	18 passing
Tamper attack types tested	7 (x 300 random trials each) plus consistent rewrite
Missed attacks / false alarms	0 / 0
Real model run	qwen3:8b, 12 entries, 3 batches, verified
Gas per anchor (first run)	about 79,500 to 93,100

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3. Decisions taken along the way

- Chose the audit-trail project as the safest to finish because its core claim (tampering is detected) is provable by test and does not depend on AI model quality.
- SHA-256 everywhere so Python and Solidity agree.

- Only Merkle roots go on-chain.
- Used a local model (Ollama, qwen3:8b) because the organisation provides no Anthropic API key.
- Hardhat 2 (not 3) for familiarity; it runs on the installed Node v26.5.0 even though that version is not officially listed as supported.

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4. Not done yet

- Evaluation experiments 2 to 4 (gas versus batch size, overhead, scaling).
- Streamlit dashboard.
- Verified literature review (citations are from memory).
- Public testnet run (optional).
- Signed entries and consistency proofs (optional extensions).
- Push to a remote repository.

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5. Things to know

- The Ollama server was started by hand and keeps running until stopped (`pkill ollama`, or `brew services stop ollama` if started as a service).
- Demo logs in `research/experiments/agent_logs/` are git-ignored because each demo run overwrites them.
- `artifacts/`, `cache/` and `node_modules/` are git-ignored and are recreated by `npx hardhat compile` and `npm install`.